

Convex Polygon Area

Day 5 solution walkthrough

Competitive Programming Bootcamp

Statement excerpts:

- ▶ *“calculate the area of convex polygons”*
- ▶ *“the last vertex is connected to the first one”*

Programming interpretation:

- ▶ The input already gives vertices in polygon order.
- ▶ The shoelace formula sums each directed edge contribution.
- ▶ Twice the area is an integer, so the final answer ends in .0 or .5.

Textbook connection: CP4 7.3.3 Area of a Polygon

Core technique: Shoelace formula

- ▶ Each test case gives a polygon as ordered vertices.
- ▶ The polygon is convex, but the shoelace formula works for any simple ordered polygon.
- ▶ The answer is the area printed with .0 or .5.

- ▶ Ordered polygon vertices suggest summing edge contributions.
- ▶ Connecting the last vertex back to the first is required.
- ▶ Integer coordinates make twice the area an integer.

- ▶ Parse the input line into n coordinate pairs.
- ▶ Initialize $\text{area2} = 0$.
- ▶ For each vertex i , let $j = (i + 1) \bmod n$.
- ▶ Add $x_i y_j - y_i x_j$ to area2 .
- ▶ Take $\text{abs}(\text{area2})$, divide by 2, and print .0 or .5.

- ▶ `area2` is twice the signed area.
- ▶ The modulo index closes the polygon without a special last-edge case.
- ▶ The sign depends on clockwise versus counterclockwise order, so take absolute value.

- ▶ The shoelace formula sums the signed area contribution of each directed polygon edge.
- ▶ Closing the polygon includes every boundary edge exactly once.
- ▶ Taking the absolute value removes orientation, yielding the geometric area.

Complexity: $O(N)$ time and $O(N)$ memory per polygon.

- ▶ You can compute while parsing to reduce memory, but storing points keeps the code clear.
- ▶ The final area is always an integer or half-integer for integer coordinates.

Source lines: `convexpolygonearea.py:41-65`

```
import sys

def main():
    input = sys.stdin.buffer.readline

    test_cases = int(input())

    for _ in range(test_cases):
        values = list(map(int, input().split()))

        n = values[0]

        points = []

        for i in range(n):
            x = values[1 + 2 * i]
            y = values[2 + 2 * i]

            points.append((x, y))
```

What this chunk does

- ▶ Each polygon is described on one line as n followed by coordinate pairs.
- ▶ The loop converts the flat coordinate list into (x, y) points.
- ▶ The points stay in the counter-clockwise order supplied by the input.

Source lines: `convexpolygonearea.py:67-84`

```
area2 = 0

for i in range(n):
    x1, y1 = points[i]

    x2, y2 = points[(i + 1) % n]

    area2 += x1 * y2 - y1 * x2

area2 = abs(area2)
```

What this chunk does

- ▶ `area2` stores twice the signed area.
- ▶ Modulo indexing closes the edge from the last vertex back to the first.
- ▶ The absolute value removes dependence on orientation.

Source lines: `convexpolygonearea.py:86-94`

```
if area2 % 2 == 0:
    print(f"{area2 // 2}.0")
else:
    print(f"{area2 // 2}.5")

main()
```

What this chunk does

- ▶ Even `area2` values produce an integer area ending in `.0`.
- ▶ Odd `area2` values produce a half-integer area ending in `.5`.
- ▶ `main()` runs the full parser and area calculation.

Solution file: `practice_contests/day_5/convexpolygonearea.py`

Problem: <https://open.kattis.com/problems/convexpolygonarea>

Use this deck with the source code open beside it. The slides explain the reasoning; the code shows the exact input handling and output formatting.